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CodexProphet

LLM forecasting for heterogeneous prediction markets

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ICML 2026 · Forecasting Workshop Oral Presentation

github.com/Hilo-Hilo/CodexProphet-ICML

Why this is hard

Why not just ask ChatGPT?

The task: one POST, one live event, 10 minutes → a probability distribution, Brier-scored against reality.

> LLMs answer from pretraining; forecasting is about *now*.

> **The gap is information, not calibration.** LLMs are calibrated out of the box — only retrieval (live prices, order books, targeted research) adds resolution.

> Your opponent is **the market**, not zero.

Our design

Design choices

Codex harness
Build on Codex, not our own agent loop

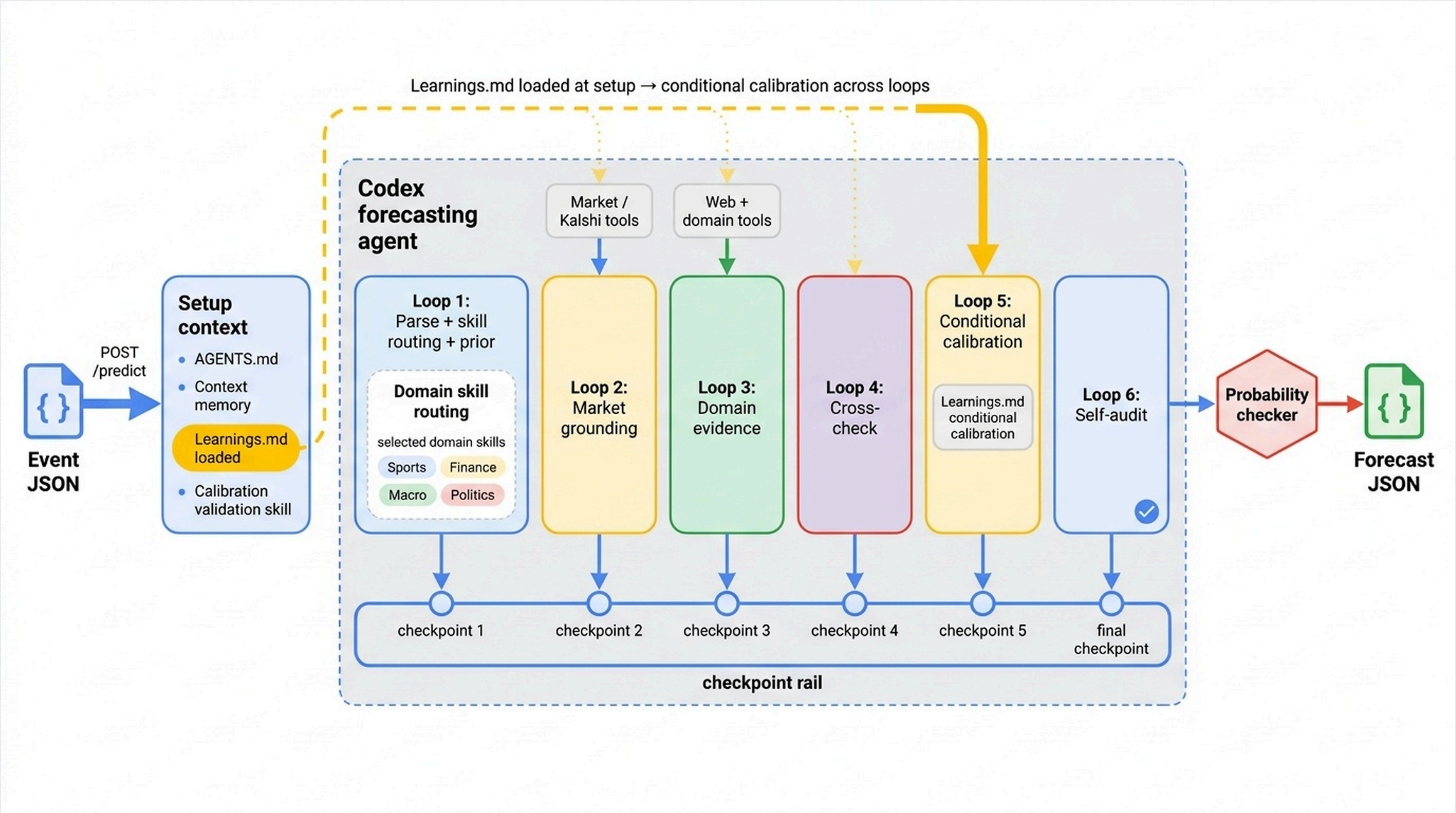
Loops
Capped at ~6, checkpointed under time limit

Skills
Curated per event category via routing

`Learnings.md`
Empirical edge from past Kalshi data

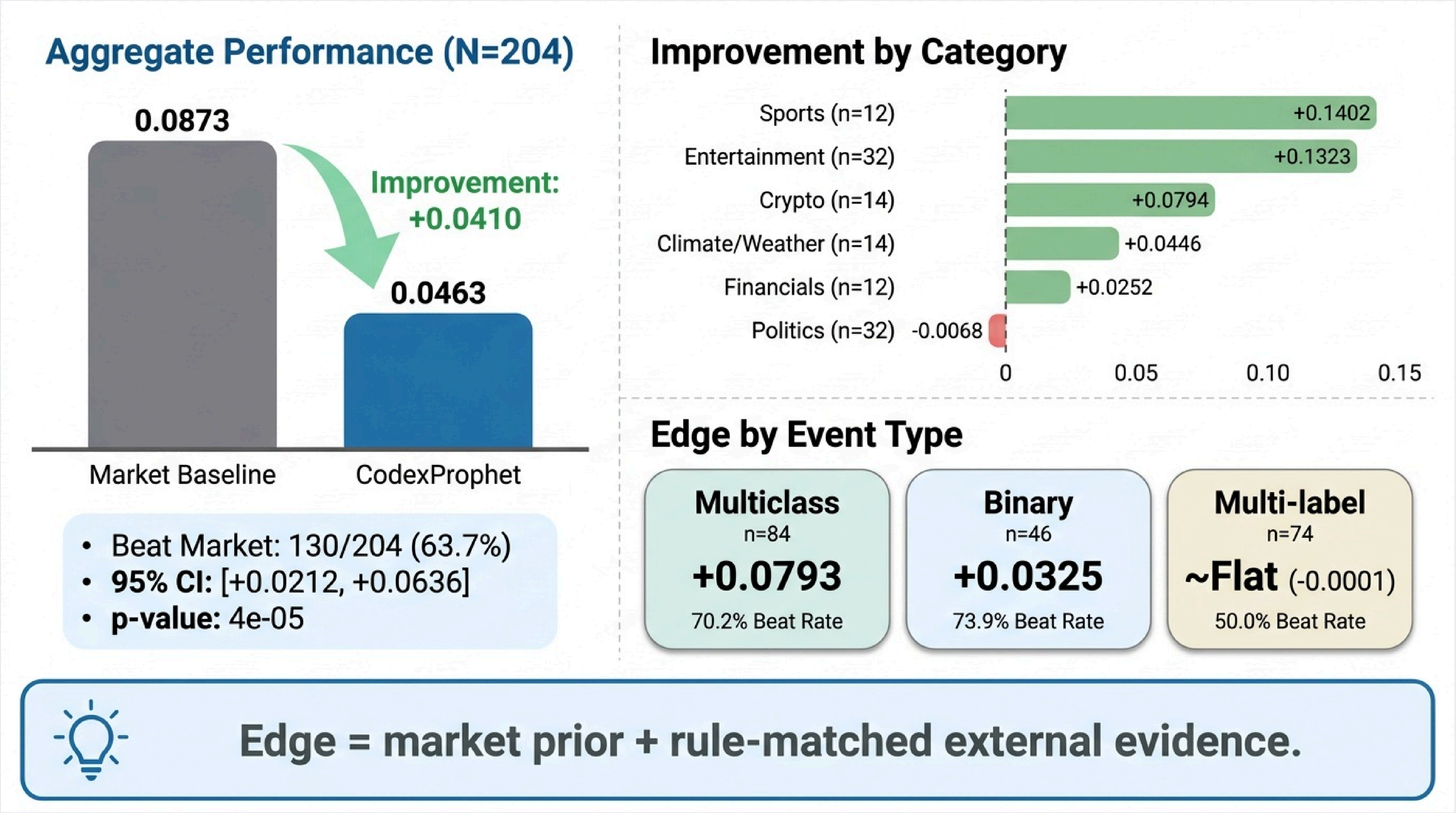
Market grounding
Anchor on live Kalshi price

Custom tools
Save time, avoid reinventing the wheel



Design overview: the CodexProphet forecasting loop. Skill-router architecture and `Learnings.md` backtest in appendix.

Hackathon result: we beat the market (+0.041, 63.7%)



Official ProphetHacks export (N = 204). Δ = market Brier – CodexProphet Brier; positive = beats the market (methodology in appendix).

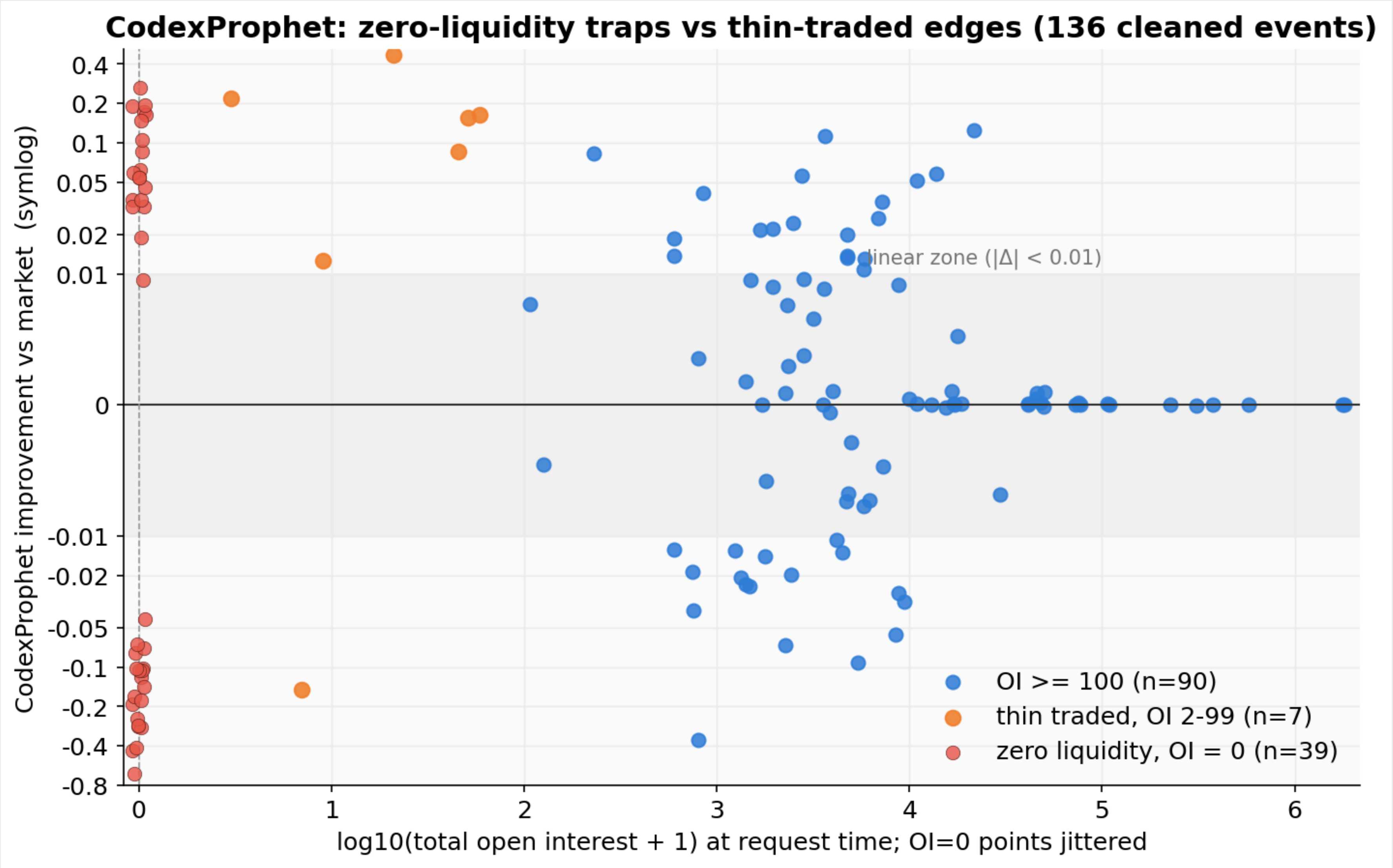
Ablation study — N = 97 real markets

Variant	Loops	Calib. skills	Market prior	Signal learn.	Market tools	Sports /fin	Median dur.	Mean Δ Brier	95% CI	Families beaten	Events beat
CodexProphet (v1) — full	✓	✓	✓	✓	✓	✓	245 s	**+0.0093**	[−0.005, +0.025]	50/71	**66%**
g — market-anchor	✗	✗	✓	✗	✓	✗	31 s	**+0.0100**	**[+0.002, +0.019]**	43/71	57%
j — adaptive	*adaptive*	✓	✓	✓	✓	✓	130 s	+0.0061	[−0.007, +0.019]	46/71	61%
l — no-skills	✓	✗	✓	✓	✓	✓	128 s	+0.0060	[−0.006, +0.019]	45/71	58%
h — no-market	✓	✓	✗	✗	✗	✓	145 s	−0.0057	[−0.032, +0.018]	39/71	51%
f — prompt-only	✗	✗	✗	✗	✗	✗	22 s	−0.0095	[−0.041, +0.018]	35/71	48%

Six arms — each removes one component of the full system. All arms use GPT-5.5 and forecast the same 97 events; zero-interest album books are excluded. Δ Brier = market Brier – arm Brier, normalized per outcome, so positive values mean the arm beats the request-time market midpoint. CIs are 20,000-resample event bootstraps; only g has a 95% CI excluding zero. Per-variant charts in appendix.

Forward eval: How does predictive power vary with liquidity?

Our own forward eval: **136 events** over one week. At first glance we lose (–0.010) — but the whole loss is one bucket: never-traded books.



Improvement vs open interest — thin-traded books (orange) hold the largest edges; zero-liquidity books (red, OI=0) drive the losses.

Bucket	Events	Beat	Δ Brier
Full set	136	61%	−0.010
Zero-interest `OI<2` 38 of 39 are album books	39	–	−0.059
Real markets `OI≥2`	97	**66%**	**+0.009**
Thin-but-traded `OI 2–99`	7	**86%**	**+0.136**
Liquid `OI≥100`	90	ties	−0.001

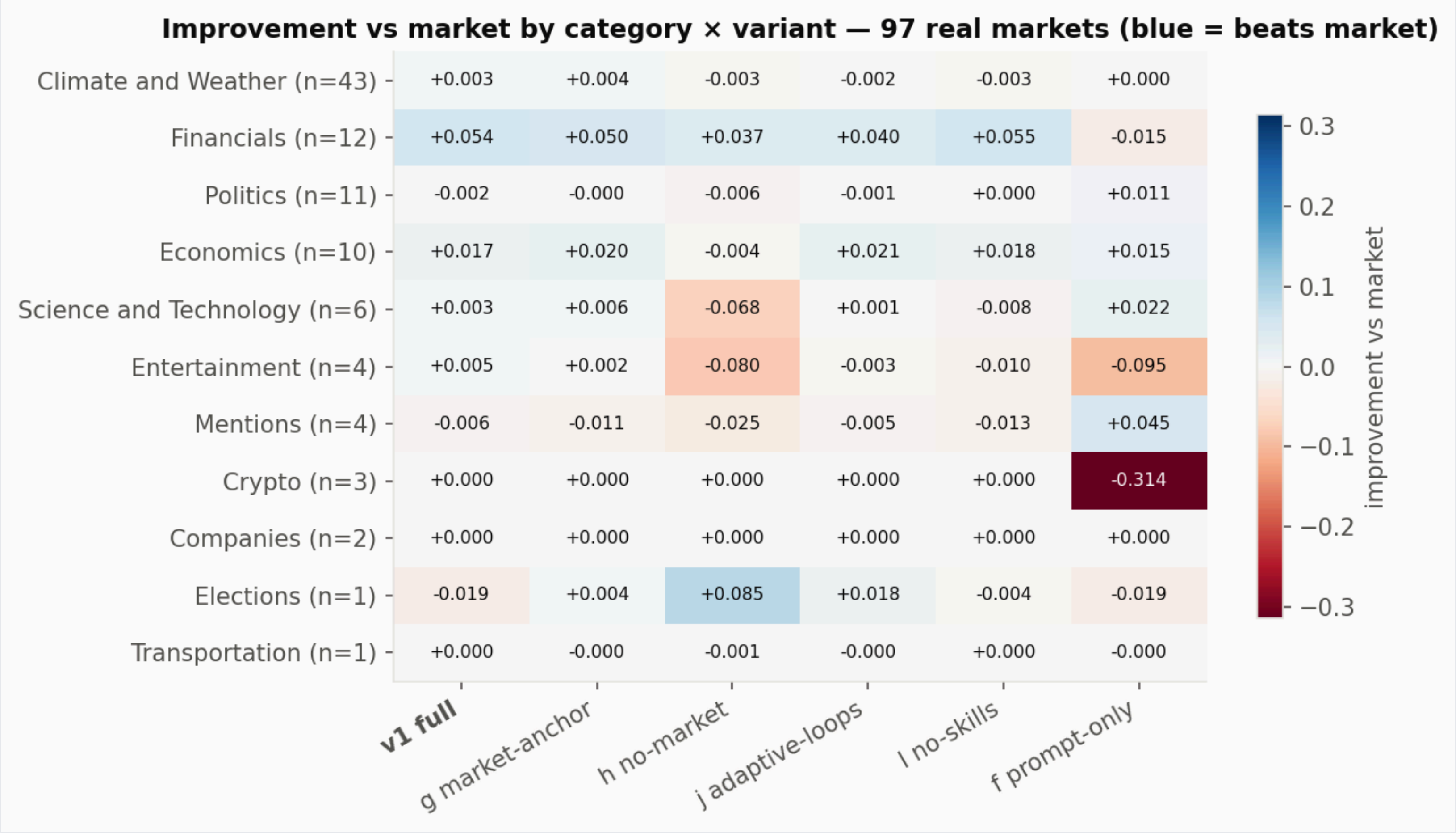
The price was fake. Empty book — one $0.96 ask, no buyers, no trades.

The research was right (correct answer: No) — but the agent anchored to the fake price and output Yes 0.94.

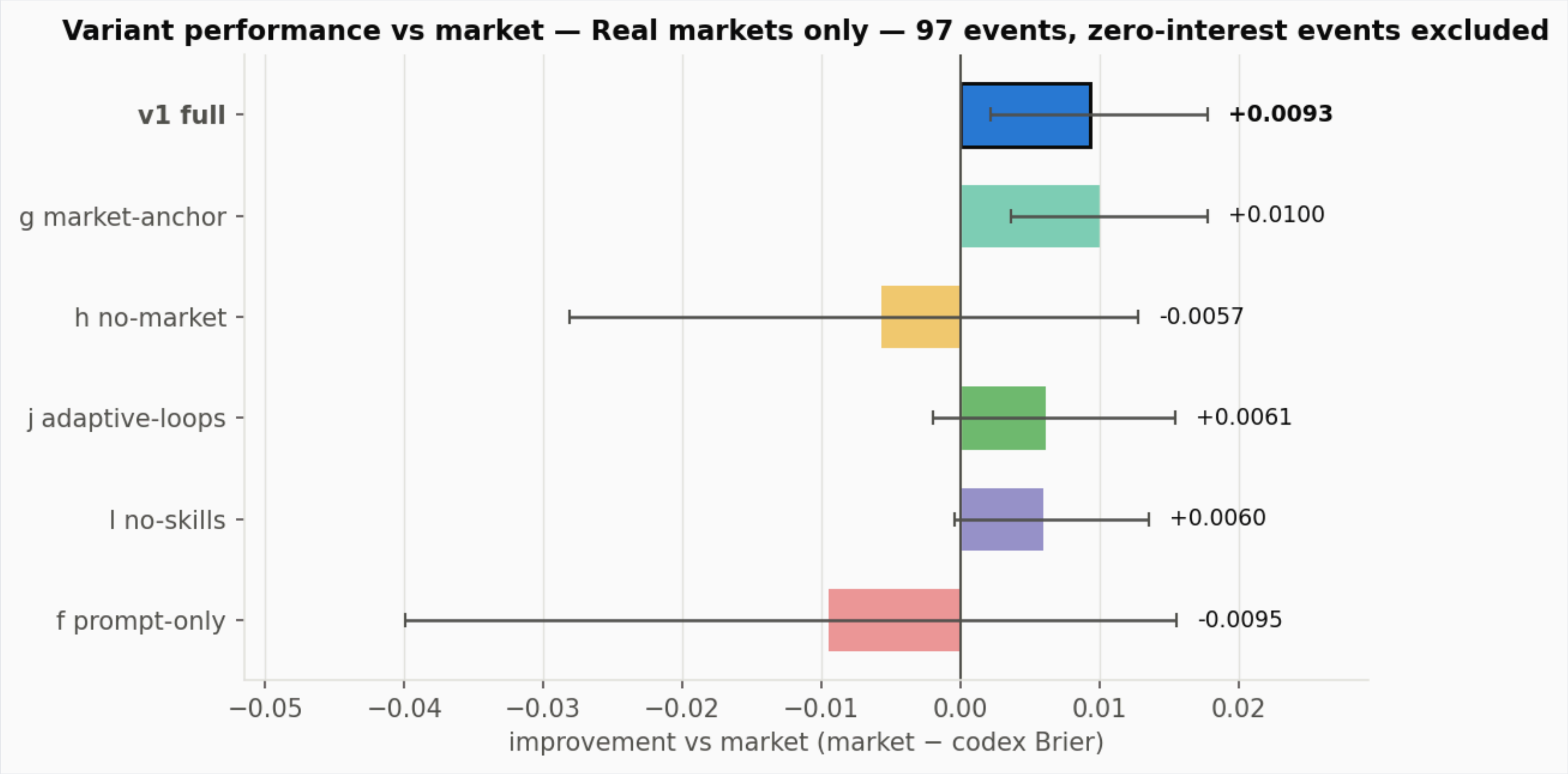
The fix: detect zero-interest books and ignore their price.

On real markets, CodexProphet beats the market again

97 real-interest events. Market-anchored variants beat the market — **v1 +0.0093 Brier, 66% of events beaten; g +0.0100 at 31 s.** Remove the market prior (h, f) and the edge disappears.



Improvement vs market by category × variant.



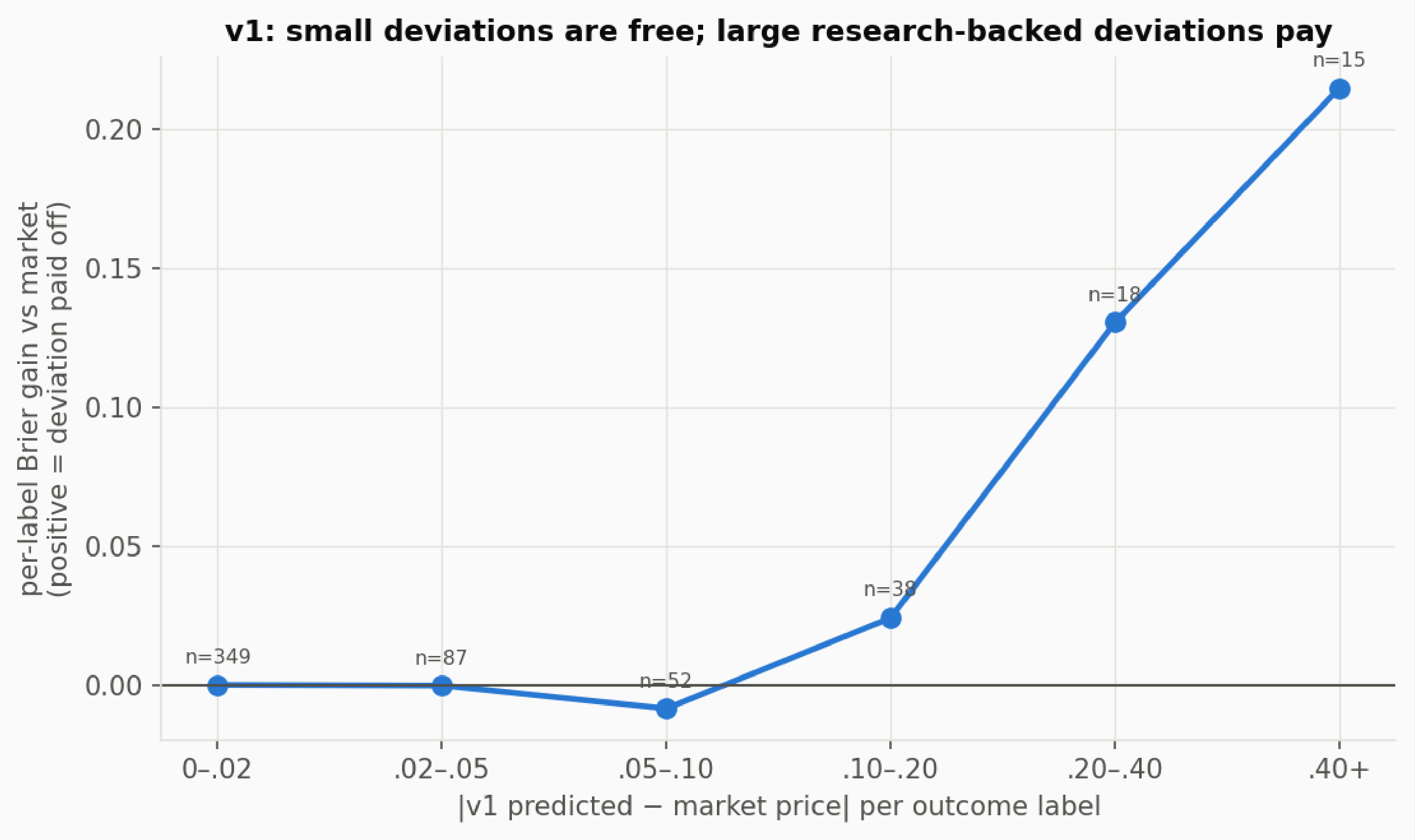
Variant performance vs market (zero-interest excluded).

Where the gain comes from

Market anchor: worth ~0.01 Brier. Remove it (h) and the edge flips negative (v1 vs h, p = 0.009).

Research loop + tools: worth ~0.04 Brier over prompt-only (v1 vs f, p = 0.034).

Research earns the right to deviate. Big researched moves (>0.15) gain +0.15–0.17 per label; the same moves without research lose –0.06 to –0.11.



Per-label Brier gain vs deviation size from the market (v1).

Findings from analysis

01 Copy the market — unless you have real evidence and the market is real.

02 We didn't build a smarter forecaster. We built a more disciplined one.

03 **Resolution, not calibration, is the differentiator:** variants differ in the information they gather, not in calibration — deleting the calibration skill (I) even improved reliability 3.4×.

FINAL THESIS

A market-aware forecasting agent that beats baseline markets when it combines **real market priors with rule-matched evidence** — and fails when it anchors on **fake liquidity**.

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github.com/Hilo-Hilo/CodexProphet-ICML

APPENDIX

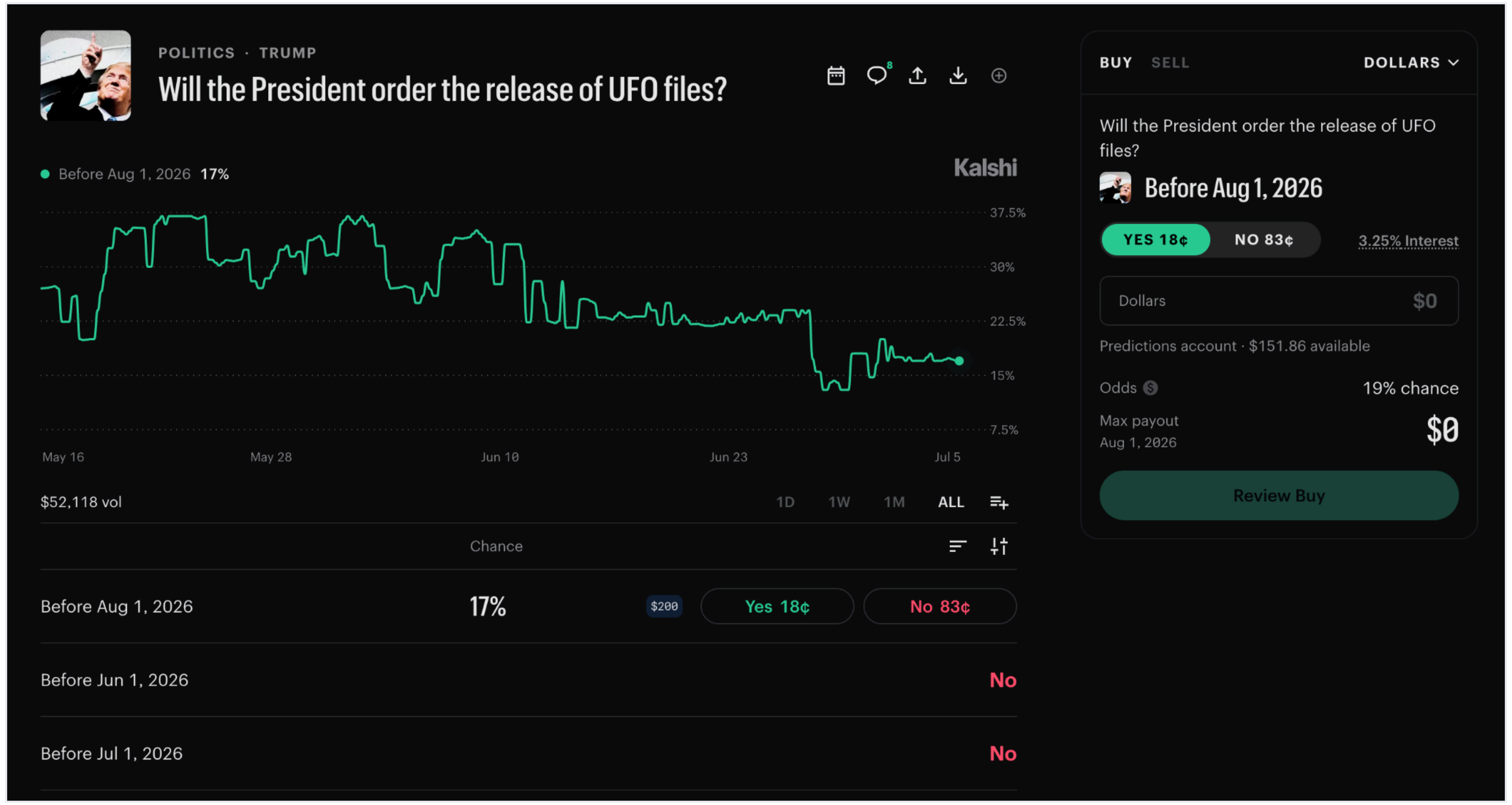
Supporting analyses for Q&A

Forecasting Kalshi market events

```
POST /predict
```

Input event title + rules + outcomes + close time + market snapshot

Output probability for every outcome, within 10 minutes



Types of events

Binary Cleveland vs Detroit

Multiclass ACM award winner

Top-k multilabel Eurovision Top 5

Threshold ladders Netflix views / CPI ranges

Competition details

Single HTTP POST to `/predict`

Payload fields

`event_ticker` `market_ticker` `title`

`category` `rules` `close_time` `outcomes[]`

Example

`KXEUROVISIONRANK-26TOP5` asks which Eurovision acts finish Top 5.

Return within 10 min

```
{ probabilities: { outcome: p },
  rationale?: string }
```

Methodology / Brier score

Brier score — the average squared error between predictions and resolutions across all labels:

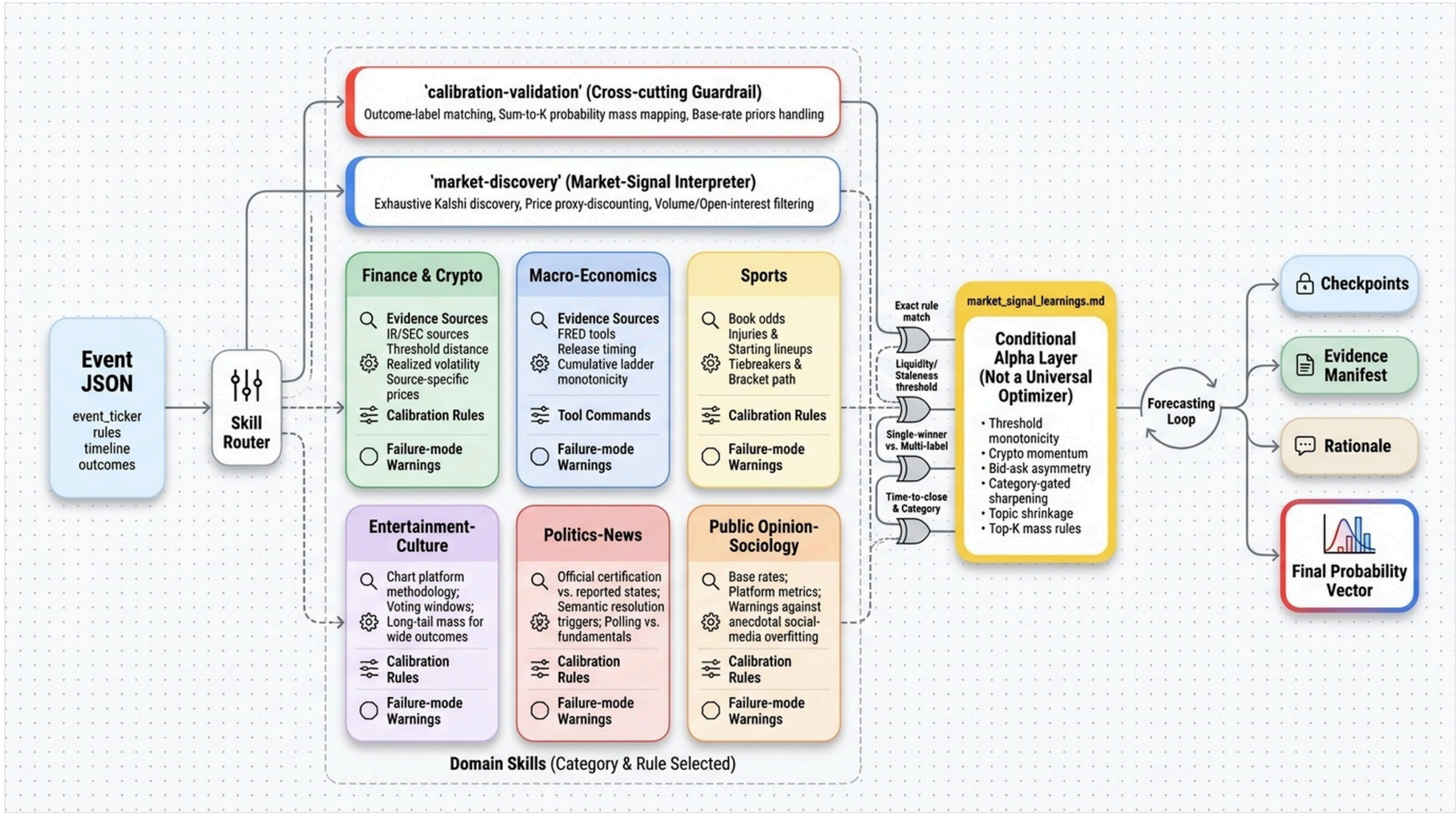
$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2$$

$$\Delta = \text{Brier}_{\text{market}} - \text{Brier}_{\text{Codex}}$$

Market comparison: the market baseline uses request-time midpoint prices. Positive Δ means the AI beat the market.

Skills and `Learnings.md`

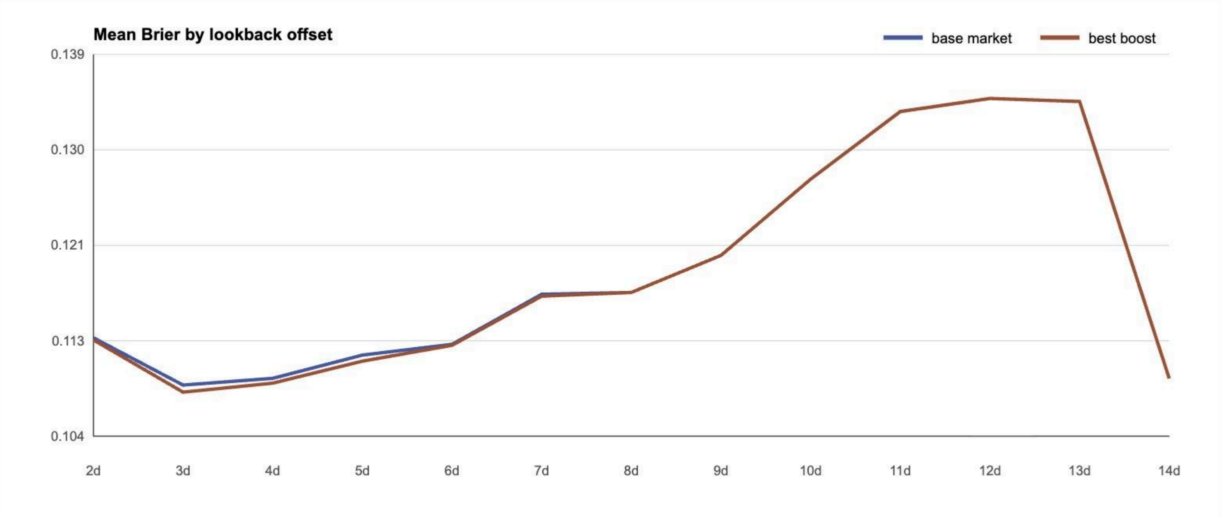
Each event routes to a curated domain skill; `Learnings.md` adds edges backtested on historical Kalshi markets.



Skill router and domain-skill architecture.

Kalshi Historical Market Backtest

Baseline uses the last available hourly Kalshi candle at or before each target time, from 48 hours to 336 hours before market close.



Best Strategy By Offset

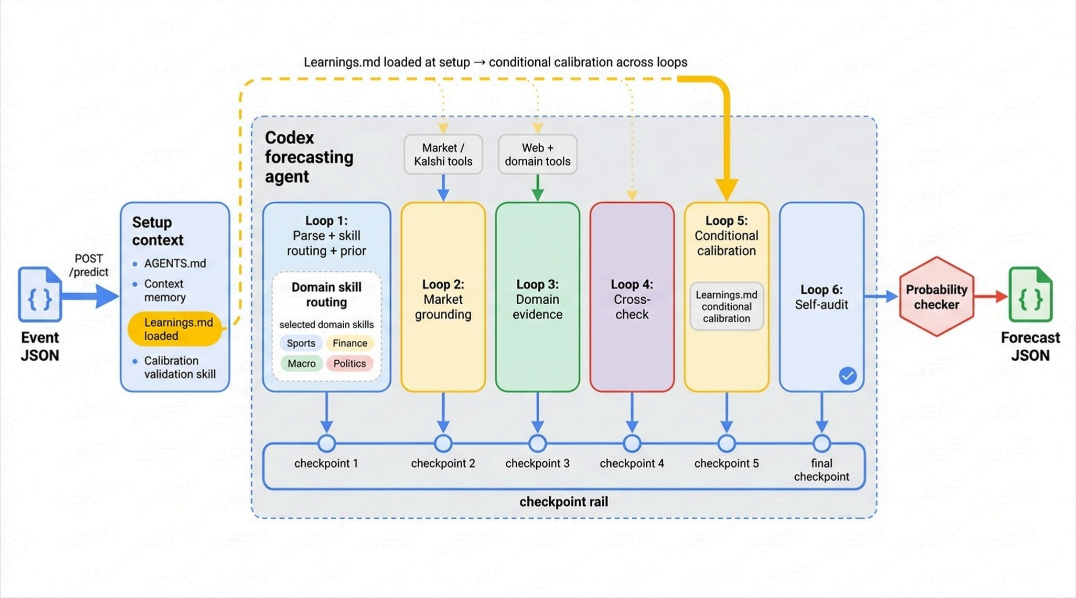
Offset hours	Strategy	Amount	Count	Mean Brier	Std
48	absolute_directional	0.02	5794	0.11273468243009571	0.17796092977934272
72	absolute_directional	0.02	5422	0.10795056252304988	0.17531850365286317
96	absolute_directional	0.02	5165	0.10877643756049922	0.17707104585243227
120	absolute_directional	0.02	4829	0.11078805653343993	0.17696850560023653
144	absolute_directional	0.01	4435	0.11224212514092217	0.18012114936261756
168	absolute_directional	0.02	3998	0.11675362056027683	0.18045130472562518
192	base	0.0	3716	0.11709494079655347	0.17594120566776397
216	base	0.0	3048	0.12049740813648213	0.18055690310544836
240	base	0.0	2750	0.12748690909090926	0.18346098412937561
264	absolute_directional	0.01	2490	0.13369437751004054	0.19115061449075918
288	base	0.0	2367	0.1348963455851295	0.19699592623548157
312	base	0.0	2170	0.13462368663594515	0.1923553177245579
336	base	0.0	344	0.10920704941860469	0.1902129702681371

Kalshi historical market backtest.

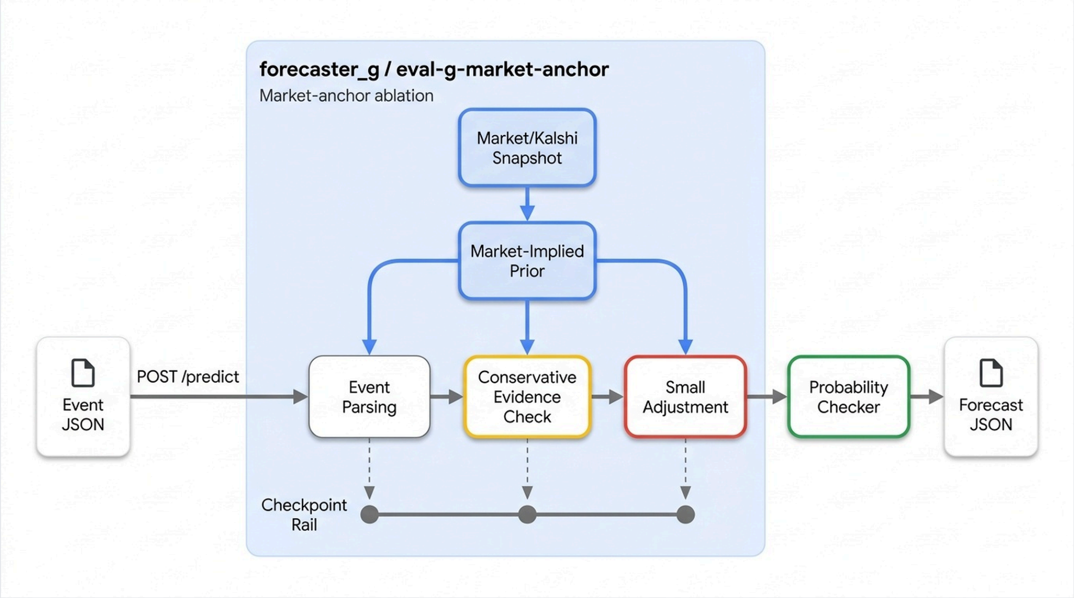
Variants

Six arms forecast the same 97 events — each removes one component of the full system.

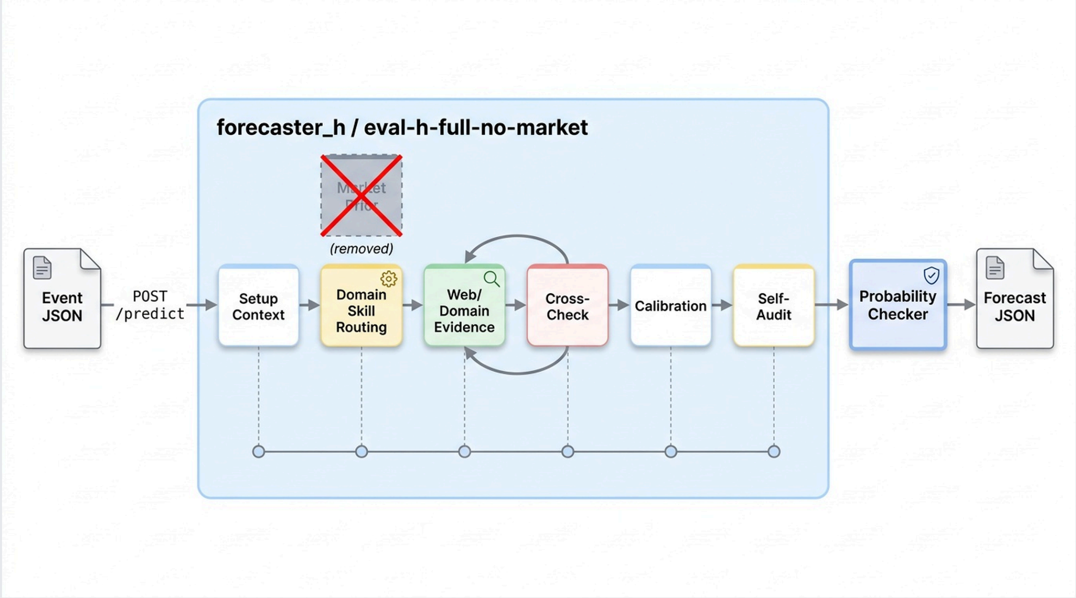
v1 · Full CodexProphet system



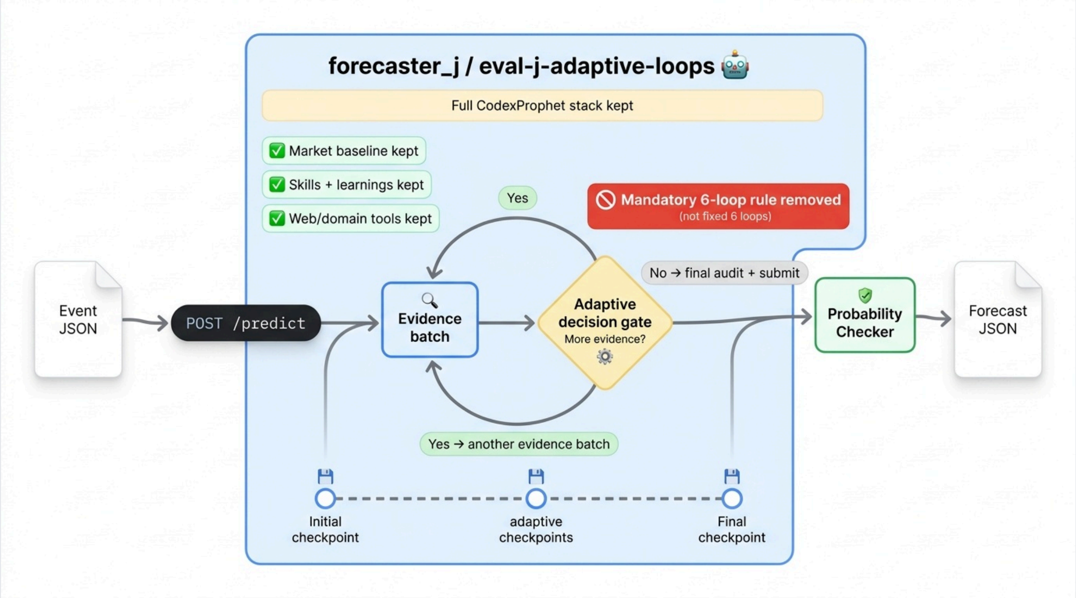
Variant G · Market-anchor



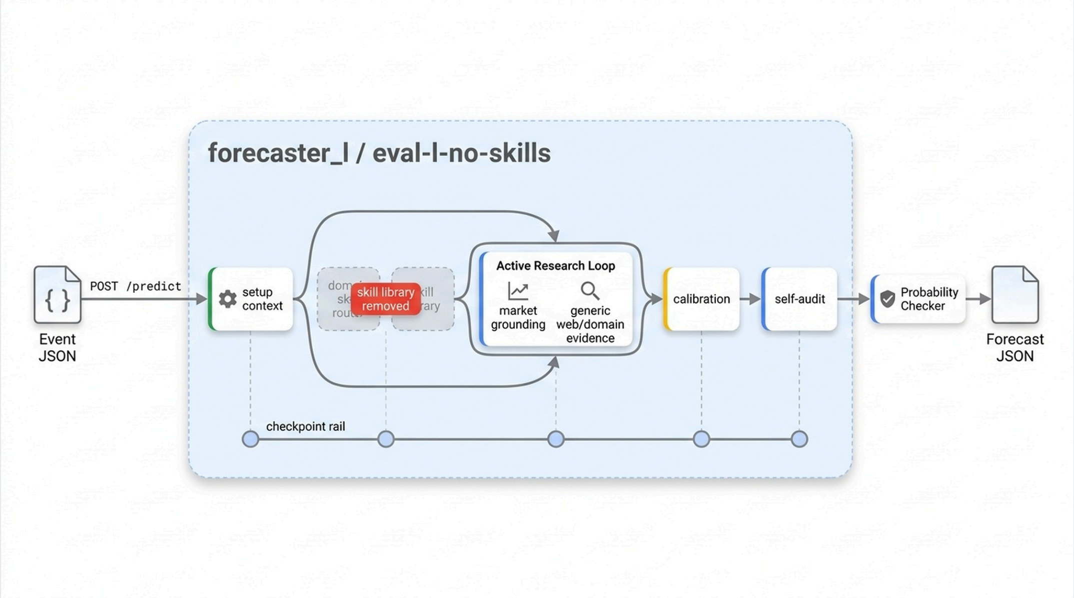
Variant H · No market grounding



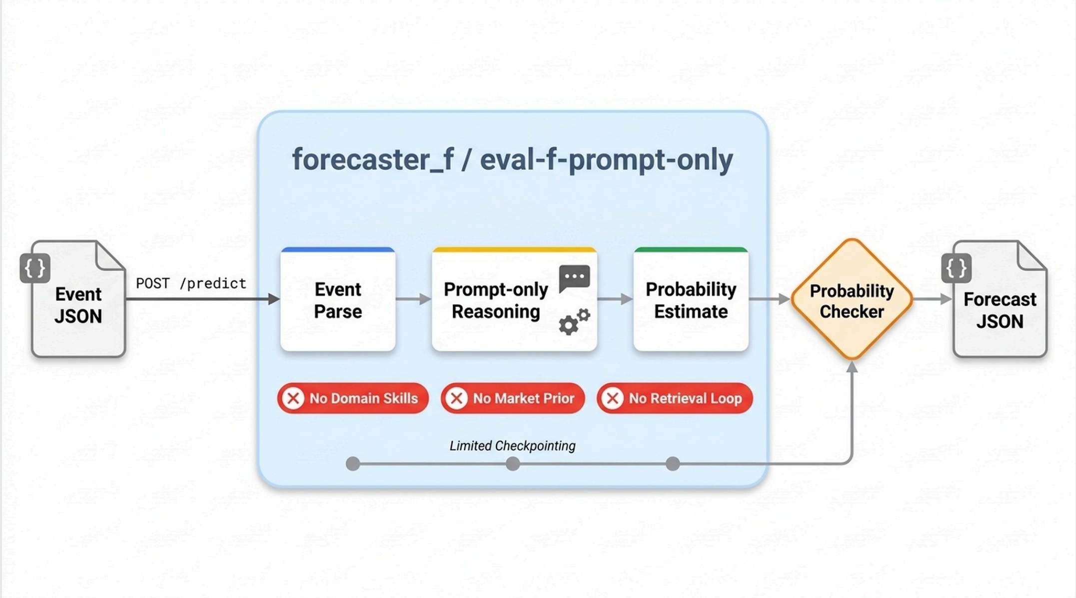
Variant J · Adaptive loops



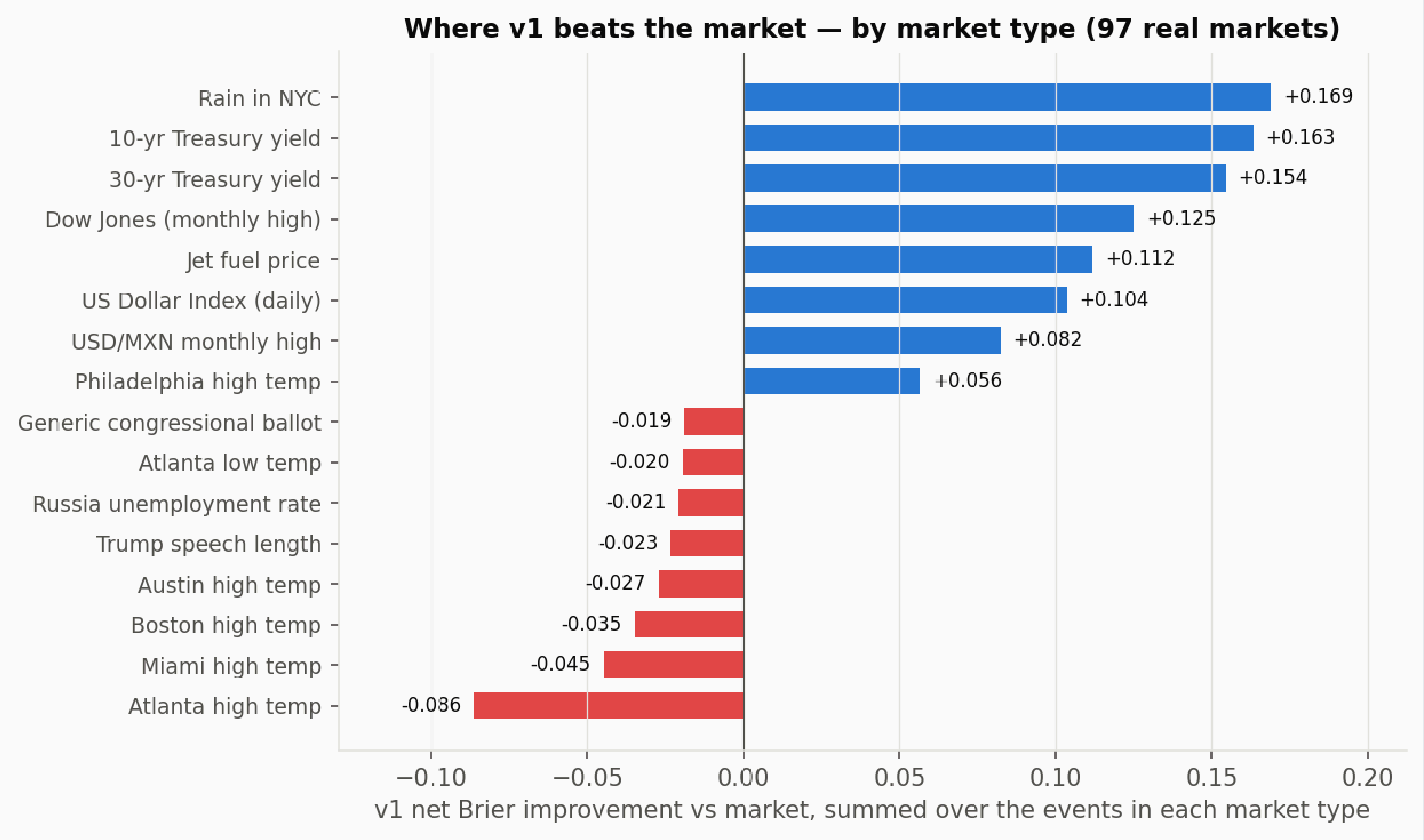
Variant L · No skills



Variant F · Prompt-only

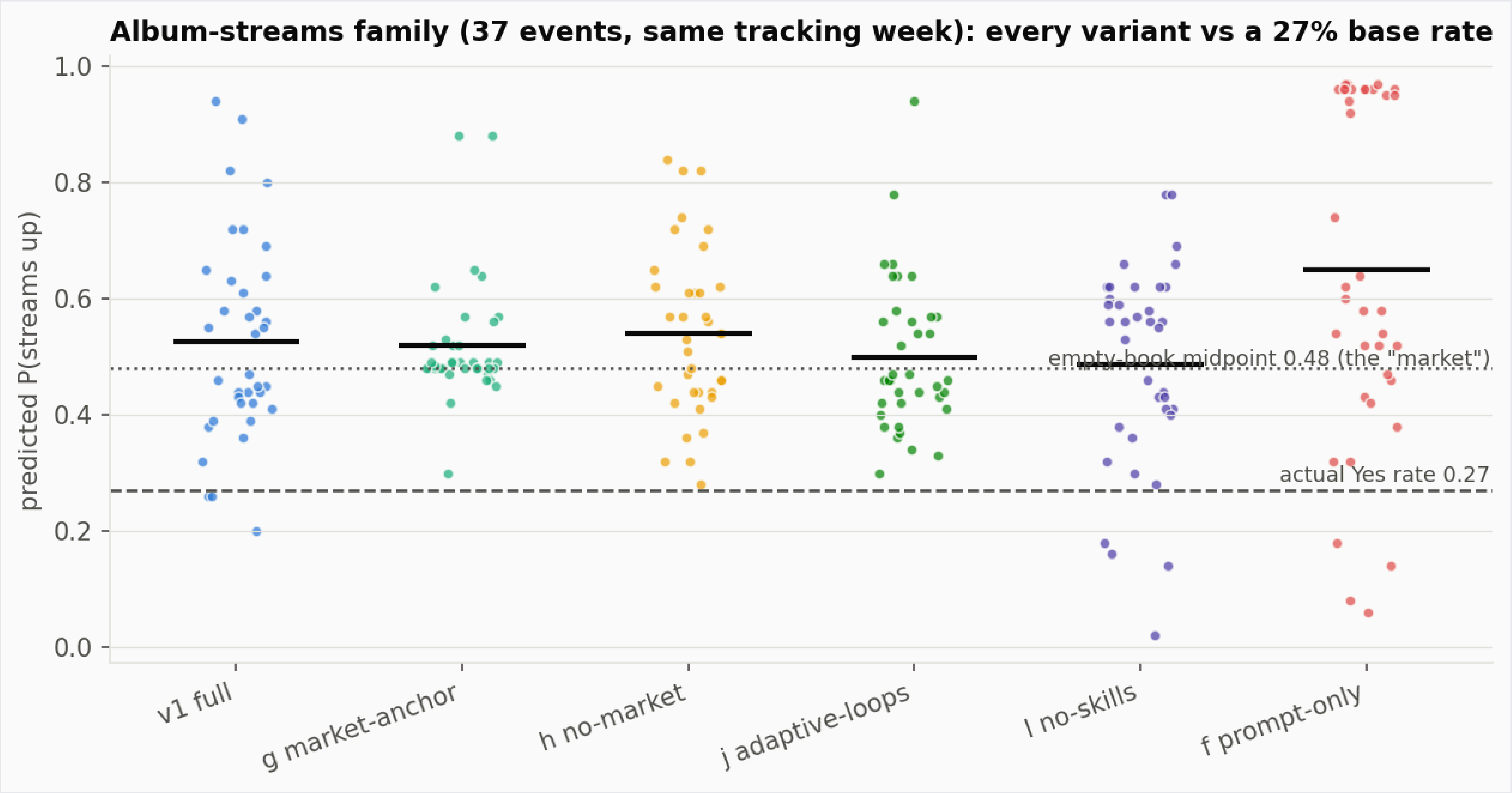


Forward eval: net improvement by market type



v1 net Brier improvement vs market, summed by market type (97 real markets).

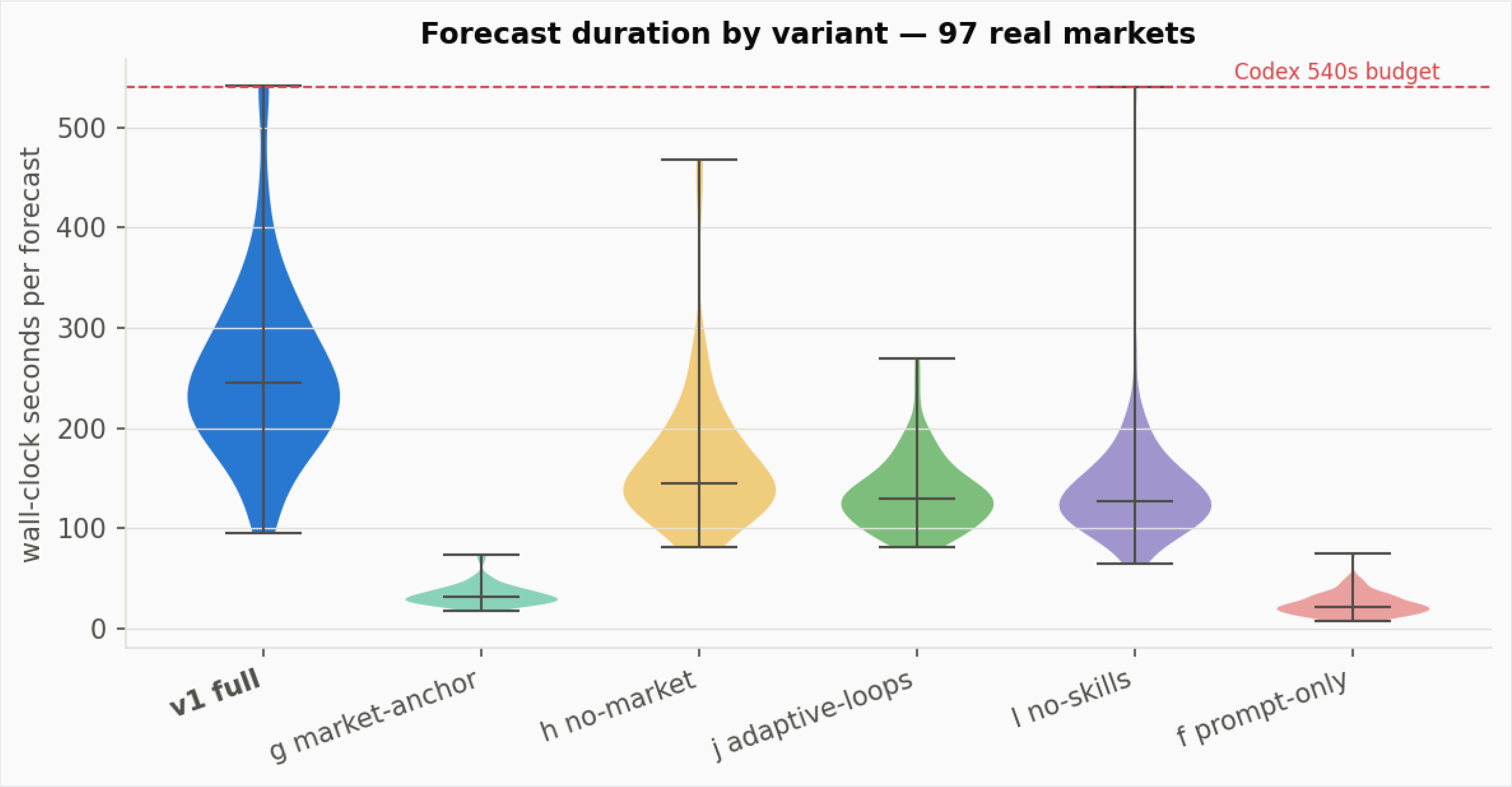
Zero-interest failure mode: album-streams family



Album-streams family: every variant vs the 27% base rate.

Compute time by variant

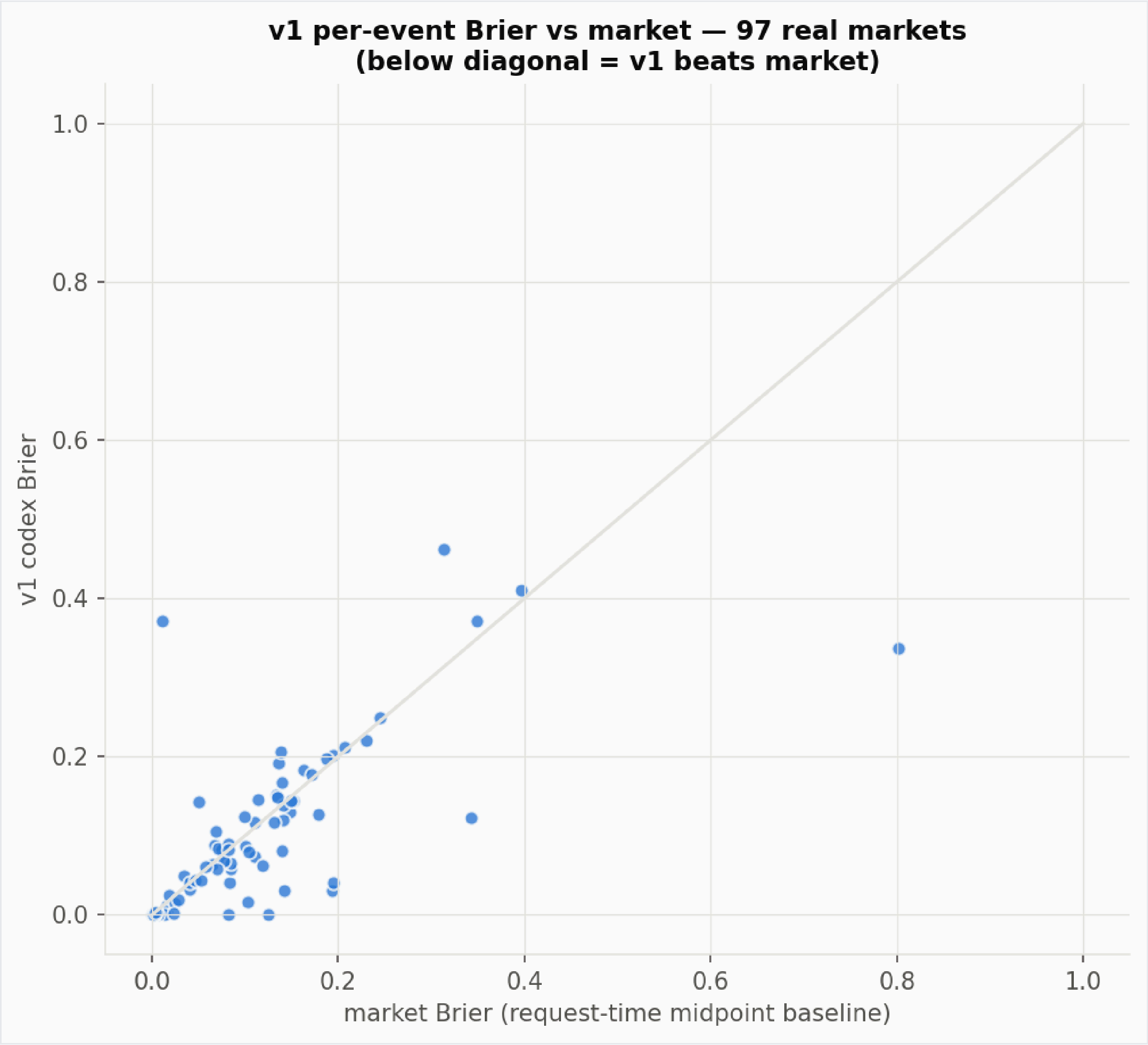
g delivers most of the edge at ~13% of the cost — 31 s vs 245 s median.



Forecast duration by variant (97 real markets).

Where the edge lives: thin-but-traded books

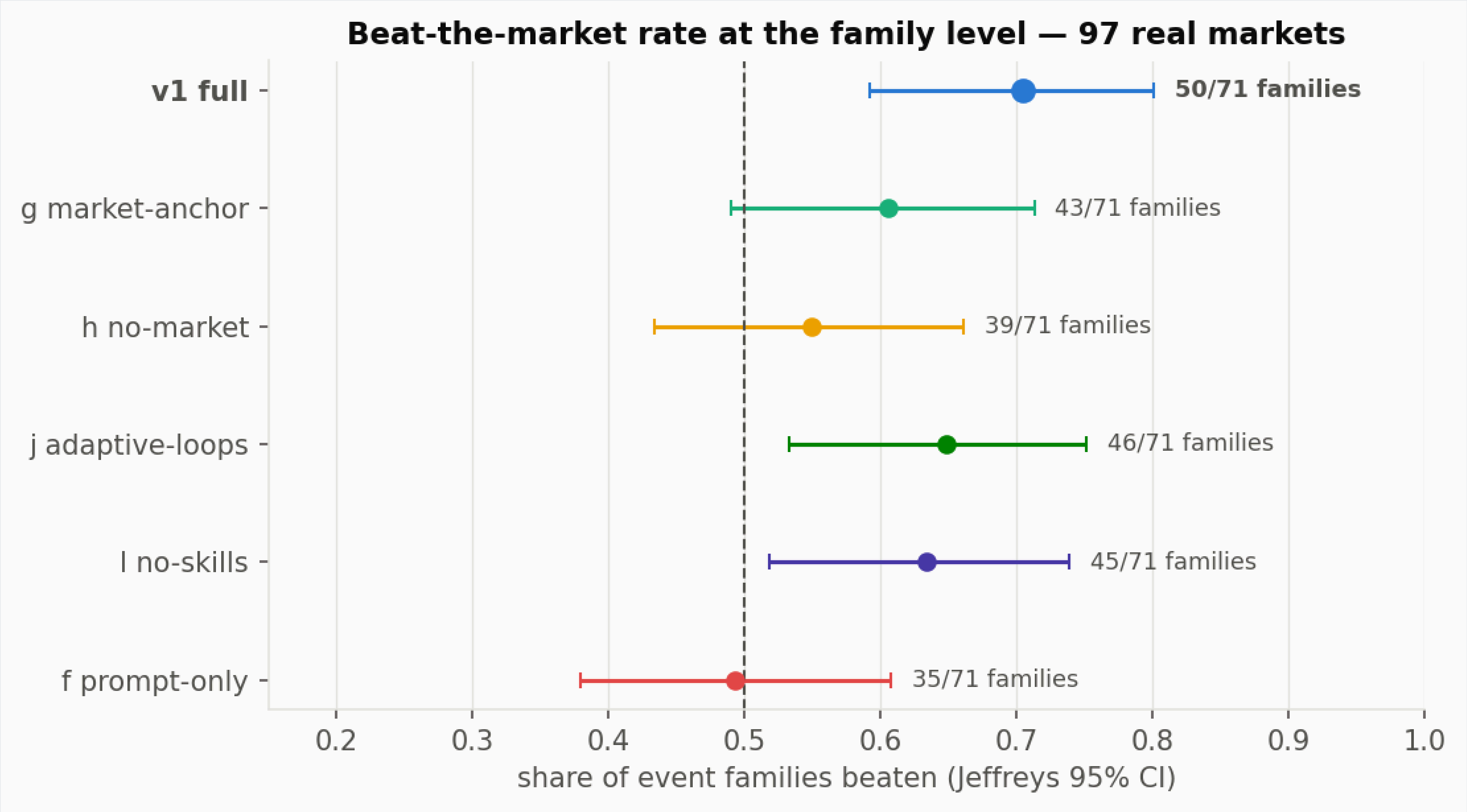
Biggest wins: thin-but-traded books with a checkable ground source — weather, yields, data calendars.



v1 per-event Brier vs market — below the diagonal = v1 beats market.

Beat rate by event family (robustness)

v1 beats 50/71 event families — the edge is broad, not a few lucky outliers.



Share of event families beaten, with Jeffreys 95% CI.

Limitations & statistical basis

Sample: 97 real-interest events · 71 event families · 6 variants (582 paired forecasts), 2026-06-29–07-05. All CIs cluster-bootstrapped by family.

Headline (real-interest set): v1 +0.0093 Brier, 66% of events beaten (50/71 families); market anchor is load-bearing — h and f (no prior) lose.

Selection effect: resolution-timing censoring over-weighted fast-resolving markets (daily weather, album streams); ~0 Sports resolved in-window — a top category in the official export. Forward-eval mix ≠ hackathon mix.

Baseline = request-time snapshot midpoint. Zero-interest album books (empty order book → no real counterfactual price) excluded as outliers; exclude-nointerest is a real-interest floor, not merely OI>0.

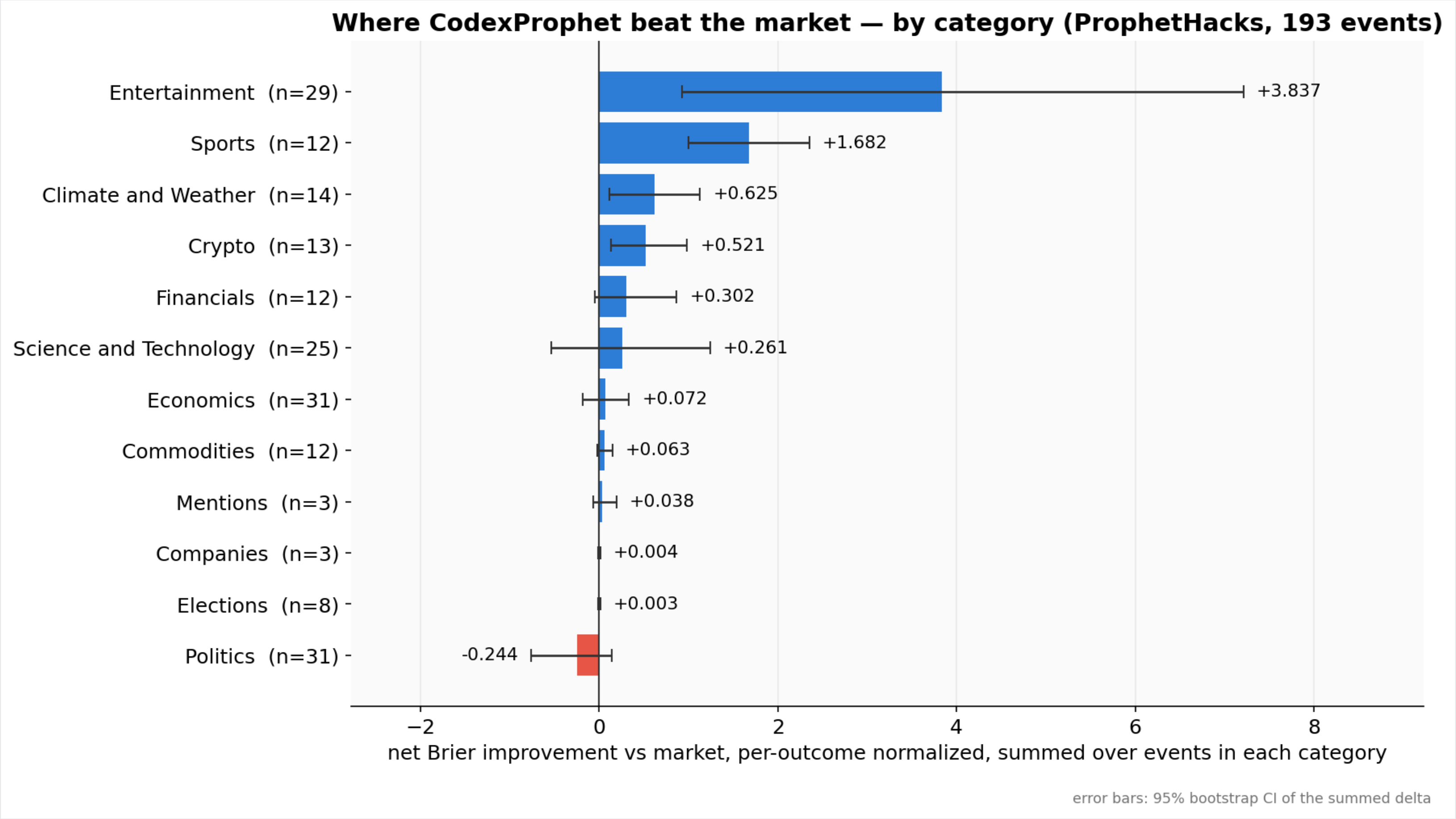
Power: ~+0.02 detectable at ~70 families; +0.01 effects need ~150. Read as directional, not definitive.

Kalshi example market payload

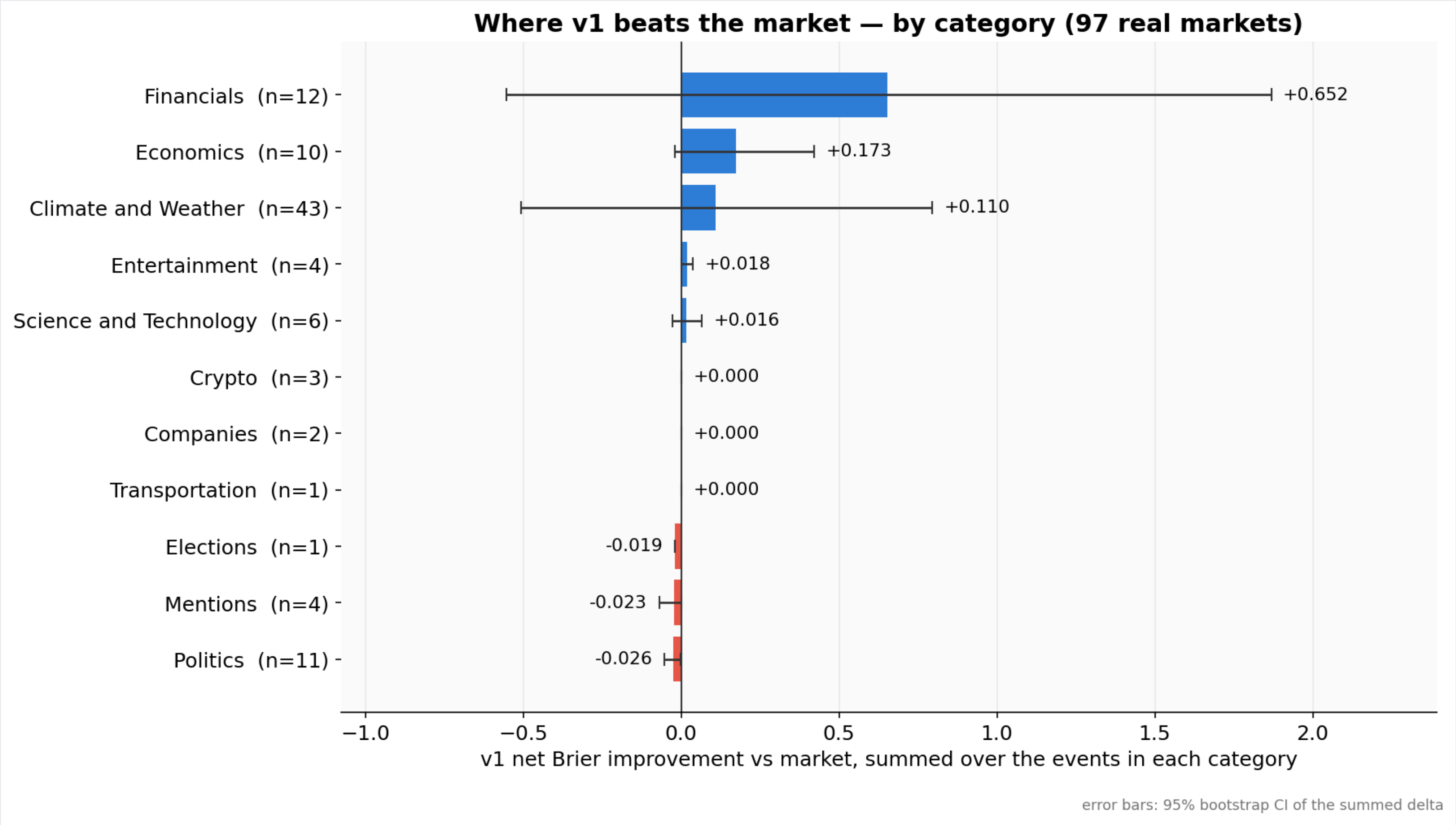
Kalshi Market Event JSON Payload


```json
{
  "event_ticker": "KXNBAGAME-26MAY15DETCLE",
  "market_ticker": "KXNBAGAME-26MAY15DETCLE",
  "title": "Will Cleveland beat Detroit in NBA Eastern Conference Game 6 on May 15, 2026?",
  "subtitle": null,
  "description": "If Cleveland wins the Game 6: Detroit at Cleveland professional basketball game originally scheduled for May 15, 2026, then the market resolves to Yes.",
  "category": "Sports",
  "rules": "If Cleveland wins the Game 6: Detroit at Cleveland professional basketball game originally scheduled for May 15, 2026, then the market resolves to Yes.",
  "close_time": "2026-05-15T20:00:00Z",
  "outcomes": ["Cleveland", "Detroit"],
  "resolved_outcome": null
}
```

Where the edge lives, by category

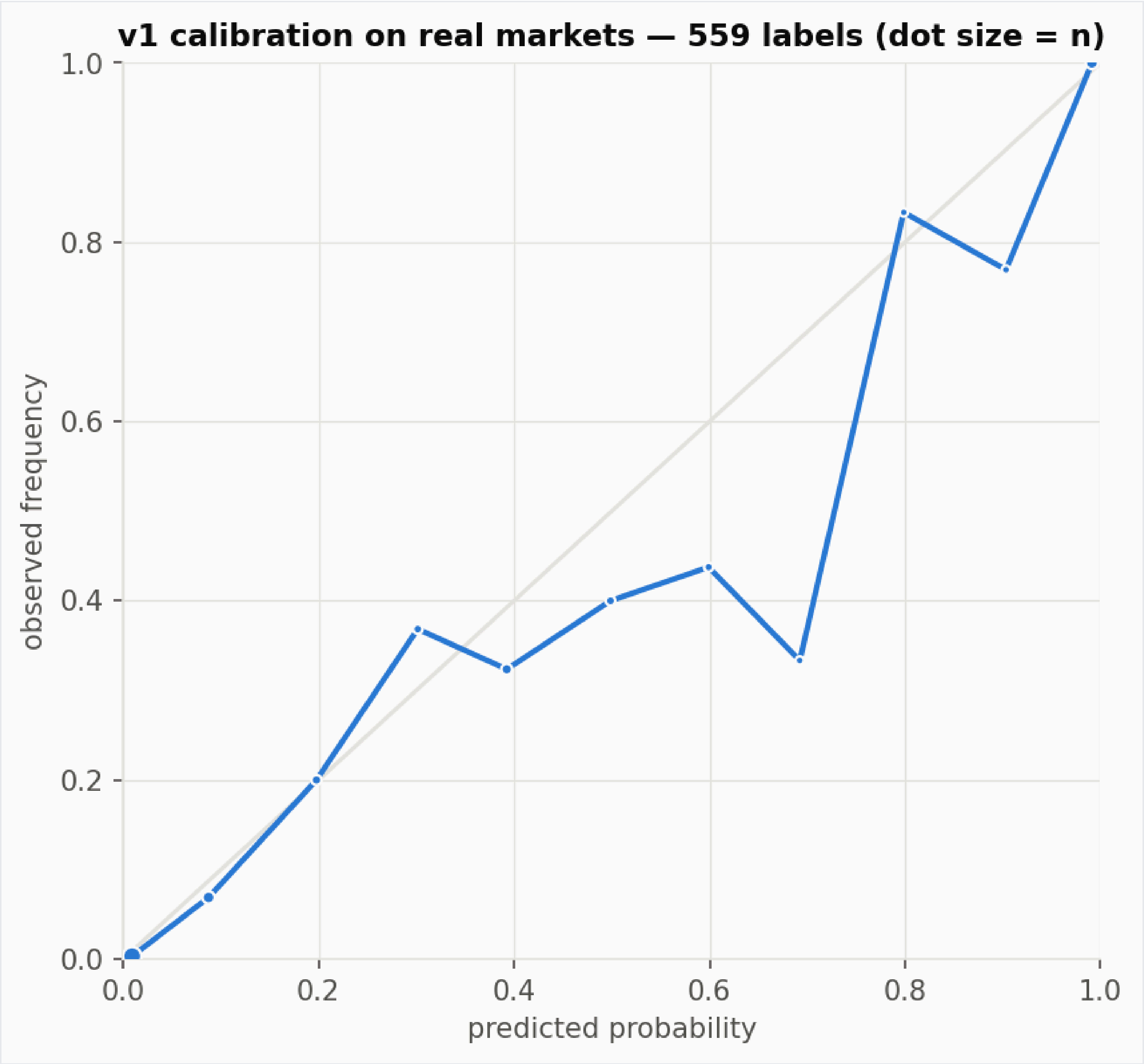


Official ProphetHacks set (193 events).



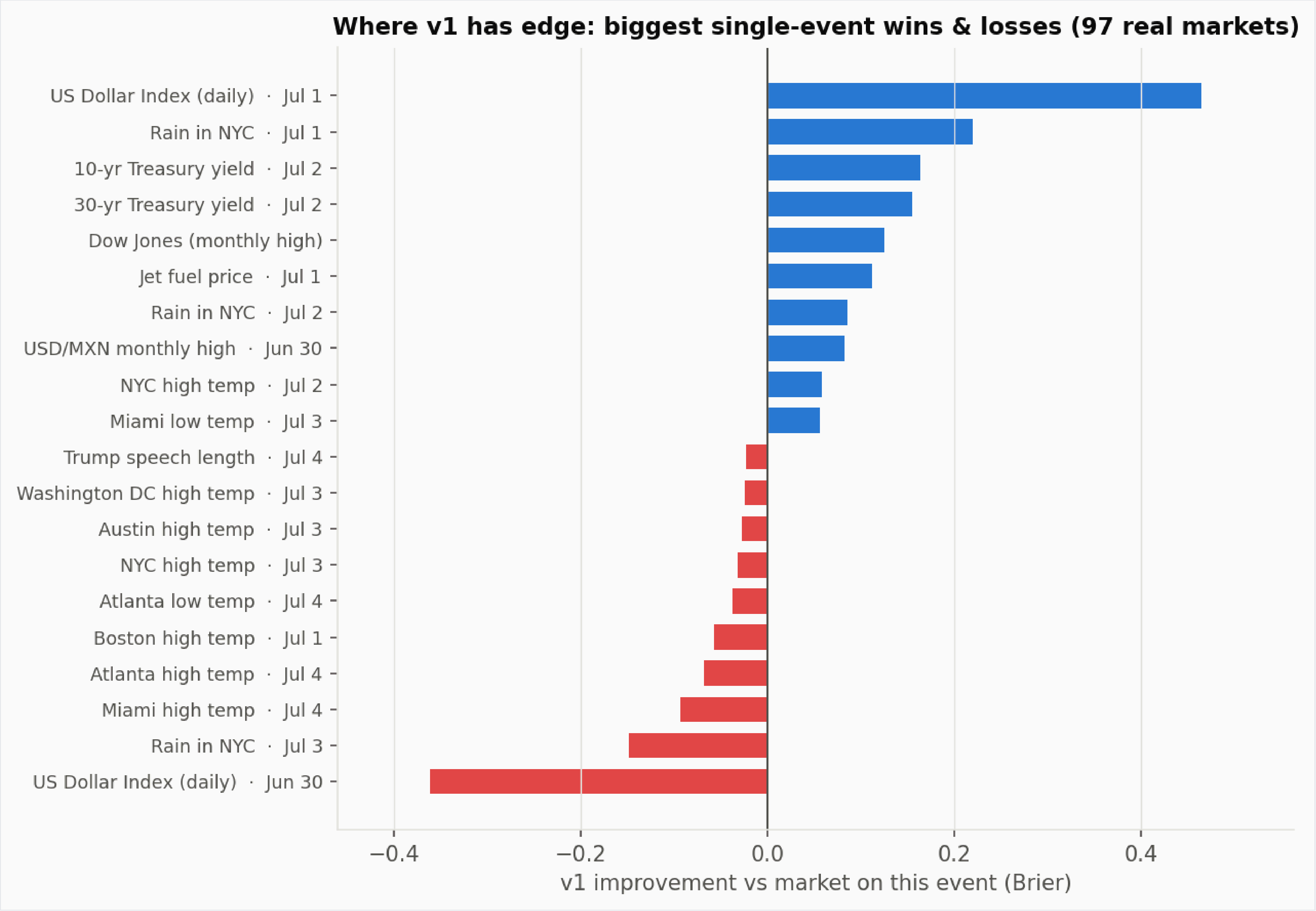
Forward-eval real-interest set (97 markets).

Calibration on real markets



Reliability curve — v1 predicted probability vs observed frequency (559 labels).

Biggest single-event wins and losses



Largest per-event Brier gains (blue) and losses (red) for v1 — 97 real markets.

How the “beat the market” score was calculated

CodexProphet	Market baseline	Improvement
`0.0463`	`0.0873`	`+0.0410`

These are **normalized post-resolution Brier errors**, not probabilities — lower is better.

Raw Brier is a vector sum across labels; non-binary events are divided by `num_outcomes` so binary and multi-outcome events compare on one scale.

Means are over the **193** of 204 rows that carry both scores.

```
normalized_brier =
  brier                    if binary
  brier / num_outcomes     otherwise
```

`0.0873` is the market’s **normalized error score — not** its prior probability.